

Informe de experimento: results_validation_gpt-5.4_prompting

Generado el 2026-09-16 09:27 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_prompting (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f ]
↪ low-agent/benchmarks/results_validation_gpt-5.4_prompting
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: single_call:cot, single_call:few_shot, single_call:rag
- Casos: case14
- Requests: generadas, N=5 por caso y seed, difficulties=todas (1 report)
- Items por dificultad: plain=6, parameterized=3, multistep=3, ambiguous=3
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 08:02
- Costo total: 0.2014 USD (15 items; costo medio 0.01343 USD/item)
- Items totales: 15; ok=15; con error=0; tokens totales=63380
- Solved global: 20.0% (3/15)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo.

Método single_call:cot

System prompt —1697 caracteres; strategy=cot; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk
```

```
## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.
```

```
## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step
```

```
## Single-call mode
You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to
↪ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
↪ and do not fabricate numbers.
```

User prompt (primer item) —4192 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
↪ index, transformers use 100000+row index.

Buses

bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)

Branches

line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)

Available tools

- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Reasoning Instructions

Before emitting tool calls, reason step by step and write a SHORT numbered plan in the message content:

1. Which buses (display ids), lines/branches and quantities/units the request refers to.
2. Which tools are needed and in which order, with the exact argument values.

Then, in the same response, emit the tool calls that implement the plan.

Output Requirements

Answer by emitting tool calls (function calling) in THIS single response:

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
↪ non-converged result, say so.

Método single_call:few_shot

System prompt —1697 caracteres; strategy=few_shot; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)

7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

Single-call mode

You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to

- call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
- and do not fabricate numbers.

User prompt (primer item) —4475 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (compact, full case)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
→ index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 7 | 14 kV
bus 8 | 12 kV | gen 0 MW vm 1.09 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
bus 11 | 0.208 kV | load 3.424 MW 1.761 Mvar
bus 12 | 0.208 kV | load 6.54 MW 1.715 Mvar
... (2 líneas omitidas de 14)
```

Branches

```
line 0 | 1-2 | r 3.532 x 10.78 ohm/km | len 1 km | max_i 42.34 kA
line 1 | 1-5 | r 9.847 x 40.65 ohm/km | len 1 km | max_i 42.34 kA
line 2 | 2-3 | r 8.564 x 36.08 ohm/km | len 1 km | max_i 42.34 kA
line 3 | 2-4 | r 10.59 x 32.13 ohm/km | len 1 km | max_i 42.34 kA
line 4 | 2-5 | r 10.38 x 31.69 ohm/km | len 1 km | max_i 42.34 kA
line 5 | 3-4 | r 12.21 x 31.17 ohm/km | len 1 km | max_i 42.34 kA
line 6 | 4-5 | r 2.433 x 7.675 ohm/km | len 1 km | max_i 42.34 kA
line 7 | 6-11 | r 4.109e-05 x 8.605e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 8 | 6-12 | r 5.318e-05 x 0.0001107 ohm/km | len 1 km | max_i 2.748e+04 kA
line 9 | 6-13 | r 2.862e-05 x 5.636e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 10 | 9-10 | r 1.376e-05 x 3.656e-05 ohm/km | len 1 km | max_i 2.748e+04 kA
line 11 | 9-14 | r 5.499e-05 x 0.000117 ohm/km | len 1 km | max_i 2.748e+04 kA
... (8 líneas omitidas de 20)
```

Available tools

- load_case(case_name): Load IEEE test case.
- run_powerflow(no arguments): Run power flow on current case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.
- get_most_loaded_branch(no arguments): Get most loaded branch.
- run_n1_contingency(top_k, criteria, max_candidates): Run N-1 contingency analysis.
- recommend_remedial_actions(max_actions, allow_load_shed, allow_voltage_control, include_best_comparison): Recommend remedial actions.
- apply_remedial_action(action_index, confirmed): Apply one remedial action by index.
- get_status(no arguments): Get current network status.
- generate_plot(plot_type): Generate one plot.

Worked Examples

The examples below refer to a DIFFERENT, fictional small system, not the case in this prompt; they show which tools to call, with which
→ arguments and in which order.

Example 1

Request: Disconnect the line between bus 2 and bus 4 and tell me which branch is now the most loaded.

Correct tool calls, in order:

1. disconnect_line(from_bus=2, to_bus=4)
2. get_most_loaded_branch()

Example 2

Request: Set the load at bus 5 to 40 MW and 10 Mvar, then rank the 3 worst N-1 contingencies by overload.

Correct tool calls, in order:

```
1. modify_load(bus_id=5, p_mw=40, q_mvar=10)
2. run_n1_contingency(top_k=3, criteria="max_overload")
```

Task

Load case14 and disconnect the line between bus ten and bus eleven.

Output Requirements

Answer by emitting tool calls (function calling) in THIS single response:

- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a
 ↳ non-converged result, say so.

Método single_call:rag

System prompt —1697 caracteres; strategy=rag; la red ya está cargada (preload_case)

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

What you can do

You complete tasks by calling tool functions:

1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

Key rules

- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

Reply format

After each analysis, reply in this structure:

1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
3. Violations, if any
4. Suggested next step

Single-call mode

You are evaluated in single-call mode: the network named in the user message is already loaded. Read the request, decide which tools to
 ↳ call and with which arguments, and emit all tool calls in this one response. Do not ask clarifying questions, do not plan across turns,
 ↳ and do not fabricate numbers.

User prompt (primer item) —1699 caracteres; case=case14, seed=1911784258, k=1, request_id=case14-ambiguous-003-s0

The network case14 is loaded. Use the tools to fulfil the request below.

System Data (retrieved rows relevant to the request; 8 of 34 rows)

Case case14: 14 buses, 15 lines, 5 transformers, total load 253.8 MW. Bus ids are IEEE display ids (1..N); line_id is the net.line row
 ↳ index, transformers use 100000+row index.

Buses

```
bus 1 | 135 kV | slack/reference vm 1.06 pu
bus 2 | 135 kV | load 21.47 MW 12.57 Mvar | gen 41.63 MW vm 1.045 pu
bus 3 | 135 kV | load 85.73 MW 17.29 Mvar | gen 0 MW vm 1.01 pu
bus 4 | 135 kV | load 45.57 MW -3.718 Mvar
bus 5 | 135 kV | load 6.887 MW 1.45 Mvar
bus 6 | 0.208 kV | load 11.74 MW 7.862 Mvar | gen 0 MW vm 1.07 pu
bus 9 | 0.208 kV | load 32.23 MW 18.14 Mvar
bus 10 | 0.208 kV | load 9.715 MW 6.261 Mvar
```

Branches

Available tools

- disconnect_line(from_bus, to_bus): Disconnect line/branch between two buses and recompute.
- load_case(case_name): Load IEEE test case.
- modify_load(bus_id, p_mw, q_mvar): Modify bus load and recompute.
- reconnect_line(from_bus, to_bus): Reconnect line/branch between two buses and recompute.

Task

Load case14 and disconnect the line between bus ten and bus eleven.

```
## Output Requirements
Answer by emitting tool calls (function calling) in THIS single response:
- Use only the provided tools, with their exact names and argument names.
- Emit every tool call the request needs, in execution order, in one round; there is no second turn and no memory of earlier turns.
- bus_id / from_bus / to_bus are IEEE display ids (1..N), as printed in the system data.
- Never invent numerical results; numbers come only from tool outputs.
- If the request needs a case other than the loaded one, call load_case first.
```

FINAL_ANSWER_INSTRUCTION (ronda final) —158 caracteres; se añade como mensaje de usuario antes de la respuesta final

Write the final answer for the user strictly from the tool outputs above. Never invent numbers; if a tool reported an error or a ↪ non-converged result, say so.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion` **utilidad de tarea** Form. (formulación exacta de las tool calls), V_MAE (p.u.) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (números de la respuesta trazables a salidas de tools) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Abst. (abstención en items resolubles) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case	N	Form.	V_MAE	F_MAE	Solved	Solver status	B_mean	Faith.	SFR	Calls	Tok.	Claim	Abst.	V pass	\$
single_call:cot	case14	5	20.0 (1/5)	4.33e-03	1.58 (1/5)	20.0 (1/5)	100.0 (5/5)	4.68e-03	100.0	–	1.0	5049	–	0.0 (0/5)	100.0 (5/5)	0.0831
single_call:few_shot	case14	5	20.0 (1/5)	4.33e-03	1.58 (1/5)	20.0 (1/5)	100.0 (5/5)	4.68e-03	100.0	–	1.0	4997	–	0.0 (0/5)	100.0 (5/5)	0.0744
single_call:rag	case14	5	20.0 (1/5)	4.33e-03	1.58 (1/5)	20.0 (1/5)	100.0 (5/5)	4.68e-03	100.0	–	1.0	2630	–	0.0 (0/5)	100.0 (5/5)	0.0439

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	plain	parameterized	multistep	ambiguous
single_call:cot	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
single_call:few_shot	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
single_call:rag	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)

Solved por dificultad (% , items resueltos / items)

Method	plain	parameterized	multistep	ambiguous
single_call:cot	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
single_call:few_shot	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)
single_call:rag	50.0 (1/2)	0.0 (0/1)	0.0 (0/1)	0.0 (0/1)

Tipos de error de formulación (conteo de items)

Method	missed_step	ok
single_call:cot	4	1
single_call:few_shot	4	1
single_call:rag	4	1

Errores de ejecución por método (conteo de items con error)

(ningún item con error)

4. Trazas de ejemplo

Hasta 3 items por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (claimed_success_on_failure, stale_state, abstained). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de `traces/` cuando existe.

Método single_call:cot

Ejemplo: resuelto (solved=True)

```
request (case14-plain-000-s0, case14, difficulty=plain, seed=1813382119): Load case14 and run the power flow, then report the bus with the
↳ lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
LLM: 1. The request refers to IEEE display bus ids 1..14 in case14 and asks for the post-power-flow bus with the lowest voltage magnitude,
↳ reported in per unit; no specific branch outage or load change is requested.
2. Needed tools and order: first `load_case(case_name="case14")` to ensure the correct ne ...[157 chars más]
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary: I loaded case14 successfully, but no power-flow or status output was provided, so the lowest-voltage bus
↳ cannot be determined from the available tool results.

2. Key numbers: Loaded case14 with 14 buses, 5 generators, 20 lines, 11 loads, total load 252.6562386075145 MW, and ...[286 chars más]
--- Respuesta final ---
1. One-sentence summary: I loaded case14 successfully, but no power-flow or status output was provided, so the lowest-voltage bus cannot be
↳ determined from the available tool results.

2. Key numbers: Loaded case14 with 14 buses, 5 generators, 20 lines, 11 loads, total load 252.6562386075145 MW, and total generation
↳ capacity 772.4000000000001 MW.

3. Violations, if any: No power-flow results were returned, so no voltage range or violations can be reported.

4. Suggested next step: Run `run_powerflow()` and then `get_status()` to identify the bus with the lowest voltage magnitude.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4969;
↳ cost_usd=0.0159; wall=5.8 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
LLM: 1. The request refers to IEEE case14, specifically the branch between bus 10 and bus 11 (display ids 10 and 11). The needed network
↳ change is to disconnect that line/branch; quantities involved are bus ids only, with no new MW/Mvar values requested.
2. Tools needed in order: first `load_case` with ` ...[180 chars más]
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
Case14 was loaded successfully, but the requested line disconnection between bus 10 and bus 11 was not executed because no tool output for
↳ `disconnect_line(10, 11)` is available.

2. Key numbers
- Total load: 253.84578240591338 MW
- Total generation capacity: 772.400000 ...[324 chars más]
--- Respuesta final ---
1. One-sentence summary
Case14 was loaded successfully, but the requested line disconnection between bus 10 and bus 11 was not executed because no tool output for
↳ `disconnect_line(10, 11)` is available.

2. Key numbers
- Total load: 253.84578240591338 MW
- Total generation capacity: 772.4000000000001 MW
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11

3. Violations, if any
No power-flow or violation results were reported in the available tool output.

4. Suggested next step
Run `disconnect_line(from_bus=10, to_bus=11)` and use its output to confirm the outage and any recomputed system results.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=2.05e-03; F_MAE=1.39; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=5009; cost_usd=0.0164; wall=8.5 s
```

Ejemplo: no resuelto

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
```

```

executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
LLM: 1. The request refers to IEEE display-id buses 6 and 13, and buses 12 and 13; the branches to open are the 6-13 line and the 12-13
  ↳ line, then rerun AC power flow and identify all branches/lines with loading above 100%.
2. Needed tools in order: `load_case(case_name="case14")`, `disconnect_line(from_ ...[158 chars más]
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
I only have a successful tool output for loading `case14`; the requested line disconnections, rerun power flow, and overloaded-line check
  ↳ were not executed in the available tool outputs.

2. Key numbers
- Case loaded: `case14`
- Buses: 14
- Generators: 5
- Lines: 20
- Loa ...[458 chars más]
--- Respuesta final ---
1. One-sentence summary
I only have a successful tool output for loading `case14`; the requested line disconnections, rerun power flow, and overloaded-line check
  ↳ were not executed in the available tool outputs.

2. Key numbers
- Case loaded: `case14`
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 257.9016499664761 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any
No post-contingency power-flow or branch-loading results are available, so I cannot list any lines above 100% loading.

4. Suggested next step
Run the remaining requested actions in order: disconnect line 6-13, disconnect line 12-13, rerun the power flow, and then query the network
  ↳ status to identify all branches with loading above 100%.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=1.96e-02; F_MAE=6.49; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
  ↳ tokens=5147; cost_usd=0.0173; wall=6.5 s

```

Método single_call:few_shot

Ejemplo: resuelto (solved=True)

```

request (case14-plain-000-s0, case14, difficulty=plain, seed=1813382119): Load case14 and run the power flow, then report the bus with the
  ↳ lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
I loaded case14, but the requested power flow was not run, so the bus with the lowest voltage magnitude cannot be reported from the
  ↳ available tool output.

2. Key numbers
- Total load: 252.6562386075145 MW
- Total generation capacity: 772.4000000000001 MW
- Total gene ...[365 chars más]
--- Respuesta final ---
1. One-sentence summary
I loaded case14, but the requested power flow was not run, so the bus with the lowest voltage magnitude cannot be reported from the
  ↳ available tool output.

2. Key numbers
- Total load: 252.6562386075145 MW
- Total generation capacity: 772.4000000000001 MW
- Total generation: not available from the tool output
- Total losses: not available from the tool output
- Voltage range: not available from the tool output

3. Violations, if any
No power-flow results are available, so violations cannot be assessed.

4. Suggested next step
Run the power flow and then query the solved bus voltages to identify the lowest-voltage bus.
--- Scoring ---

```

```
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=4967;
↳ cost_usd=0.0146; wall=6.6 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
```

```
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
```

```
executed_calls: load_case({"case_name": "case14"})
```

```
fuentes de la traza: archivo de traza completo
```

```
--- Ronda 1 ---
```

```
tool: load_case({"case_name": "case14"})
```

```
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
↳ "total_gen_capacity_mw": 772.4000000000001}
```

```
gate: n/a (no es un resultado de flujo)
```

```
--- Ronda 2 [final] ---
```

```
LLM: 1. One-sentence summary
```

```
case14 was loaded successfully, but the requested line disconnection between bus 10 and bus 11 was not executed in the available tool
```

```
↳ outputs.
```

```
2. Key numbers
```

```
- Total load: 253.84578240591338 MW
```

```
- Total generation capacity: 772.4000000000001 MW
```

```
- Buses: 14
```

```
- Generat ...[287 chars más]
```

```
--- Respuesta final ---
```

```
1. One-sentence summary
```

```
case14 was loaded successfully, but the requested line disconnection between bus 10 and bus 11 was not executed in the available tool
```

```
↳ outputs.
```

```
2. Key numbers
```

```
- Total load: 253.84578240591338 MW
```

```
- Total generation capacity: 772.4000000000001 MW
```

```
- Buses: 14
```

```
- Generators: 5
```

```
- Lines: 20
```

```
- Loads: 11
```

```
3. Violations, if any
```

```
No power-flow or post-disconnection results were returned, so violations cannot be assessed.
```

```
4. Suggested next step
```

```
Run the line disconnection for buses 10-11 and re-solve the network to report the post-contingency status.
```

```
--- Scoring ---
```

```
formulation=missed_step (exact=False); V_MAE=2.05e-03; F_MAE=1.39; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
```

```
↳ tokens=4958; cost_usd=0.0146; wall=5.4 s
```

Ejemplo: no resuelto

```
request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
```

```
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})
```

```
executed_calls: load_case({"case_name": "case14"})
```

```
fuentes de la traza: archivo de traza completo
```

```
--- Ronda 1 ---
```

```
tool: load_case({"case_name": "case14"})
```

```
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
↳ "total_gen_capacity_mw": 772.4000000000001}
```

```
gate: n/a (no es un resultado de flujo)
```

```
--- Ronda 2 [final] ---
```

```
LLM: I only have one tool output: the case was loaded successfully.
```

```
1. One-sentence summary
```

```
case14 was loaded successfully, but the requested line disconnections, rerun power flow, and overload check were not executed in the
```

```
↳ available tool outputs.
```

```
2. Key numbers
```

```
- Total load: 257.9016499664761 MW ...[414 chars más]
```

```
--- Respuesta final ---
```

```
I only have one tool output: the case was loaded successfully.
```

```
1. One-sentence summary
```

```
case14 was loaded successfully, but the requested line disconnections, rerun power flow, and overload check were not executed in the
```

```
↳ available tool outputs.
```

```
2. Key numbers
```

```
- Total load: 257.9016499664761 MW
```

```
- Total generation capacity: 772.4000000000001 MW
```

```
3. Violations, if any
```

```
No power-flow or branch-loading results were provided, so I cannot determine whether any line is above 100% loading.
```

```
4. Suggested next step
```

```
Run the remaining requested tool steps in order: disconnect line 6-13, disconnect line 12-13, rerun/obtain the solved network status, and
↳ then check branch loadings to list all lines above 100%.
```

```

--- Scoring ---
formulation=missed_step (exact=False); V_MAE=1.96e-02; F_MAE=6.49; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=5056; cost_usd=0.0151; wall=5.3 s

```

Método single_call:rag

Ejemplo: resuelto (solved=True)

```

request (case14-plain-000-s0, case14, difficulty=plain, seed=1813382119): Load case14 and run the power flow, then report the bus with the
↳ lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> run_powerflow({})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
case14 was loaded successfully, but no power-flow result was returned, so the bus with the lowest voltage magnitude cannot be determined
↳ from the available tool outputs.

2. Key numbers
- Total load: 252.6562386075145 MW
- Total generation capacity: 772.4000000000001 MW ...[392 chars más]
--- Respuesta final ---
1. One-sentence summary
case14 was loaded successfully, but no power-flow result was returned, so the bus with the lowest voltage magnitude cannot be determined
↳ from the available tool outputs.

2. Key numbers
- Total load: 252.6562386075145 MW
- Total generation capacity: 772.4000000000001 MW
- Total generation: not available from tool output
- Total losses: not available from tool output
- Voltage range: not available from tool output

3. Violations, if any
No power-flow results were returned, so violations cannot be assessed.

4. Suggested next step
Run `run_powerflow()` and then inspect the solved bus voltages to identify the bus with the lowest voltage magnitude.
--- Scoring ---
formulation=ok (exact=True); V_MAE=0; F_MAE=0.00; faithful=100.0 %; solved=True (numeric_ok); flags=-; error=-; tokens=2552;
↳ cost_usd=0.0086; wall=5.9 s

```

Ejemplo: formulación fallida (formulation_exact=False)

```

request (case14-ambiguous-003-s0, case14, difficulty=ambiguous, seed=1911784258): Load case14 and disconnect the line between bus ten and
↳ bus eleven.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11})
executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 253.84578240591338,
    ↳ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: I loaded case14 successfully.

Key numbers: 14 buses, 5 generators, 20 lines, 11 loads; total load 253.84578240591338 MW; total generation capacity 772.4000000000001 MW.

Violations: No power-flow or line-disconnection result was returned, so I cannot report the effect of disconnecting bus 10-11. No ...[181
↳ chars más]
--- Respuesta final ---
I loaded case14 successfully.

Key numbers: 14 buses, 5 generators, 20 lines, 11 loads; total load 253.84578240591338 MW; total generation capacity 772.4000000000001 MW.

Violations: No power-flow or line-disconnection result was returned, so I cannot report the effect of disconnecting bus 10-11. No error was
↳ reported for loading the case.

Suggested next step: Call `disconnect_line(from_bus=10, to_bus=11)` and use its recomputed results to report the post-outage system state.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=2.05e-03; F_MAE=1.39; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↳ tokens=2528; cost_usd=0.0081; wall=7.2 s

```

Ejemplo: no resuelto

```

request (case14-multistep-002-s0, case14, difficulty=multistep, seed=1627694679): Load the 14-bus test case (case14), disconnect the line
↳ between bus 6 and bus 13, disconnect the line between bus 12 and bus 13, and finally rerun the power flow, then list every line whose
↳ loading is above 100%.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 6, "to_bus": 13}) -> disconnect_line({"from_bus": 12,
↳ "to_bus": 13}) -> run_powerflow({})

```

```

executed_calls: load_case({"case_name": "case14"})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 257.9016499664761,
    ↪ "total_gen_capacity_mw": 772.4000000000001}
    gate: n/a (no es un resultado de flujo)
--- Ronda 2 [final] ---
LLM: 1. One-sentence summary
Loaded case14 successfully, but the requested line disconnections and final power-flow rerun were not executed in the available tool
↪ outputs.

2. Key numbers
- Case: case14
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 257.9016499664761 MW ...[369 chars más]
--- Respuesta final ---
1. One-sentence summary
Loaded case14 successfully, but the requested line disconnections and final power-flow rerun were not executed in the available tool
↪ outputs.

2. Key numbers
- Case: case14
- Buses: 14
- Generators: 5
- Lines: 20
- Loads: 11
- Total load: 257.9016499664761 MW
- Total generation capacity: 772.4000000000001 MW

3. Violations, if any
Cannot determine line loadings above 100% because no disconnection results or power-flow results were provided.

4. Suggested next step
Execute the remaining required tool calls in order: disconnect line 6-13, disconnect line 12-13, then run the power flow and inspect branch
↪ loading results.
--- Scoring ---
formulation=missed_step (exact=False); V_MAE=1.96e-02; F_MAE=6.49; faithful=100.0 %; solved=False (formulation); flags=-; error=-;
↪ tokens=2785; cost_usd=0.0094; wall=4.8 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- `V_MAE` $\geq$ 1e-4 p.u. en 3 filas método×caso con solver: `single_call:cot/case14` (4.33e-03), `single_call:few_shot/case14` (4.33e-03), `single_call:rag/case14` (4.33e-03).
- El gate no verificó ningún resultado de flujo (`gate_checked=0` en todos los items).
- Ningún item terminó con **error**.
- Solved más alto: `single_call:rag` 20.0 % (1/5); más bajo: `single_call:cot` 20.0 % (1/5).
- Formulación exacta agregada sobre métodos con tools: 20.0 % (3/15).
- Formulación por dificultad: mejor `plain` 50.0 % (3/6), peor `parameterized` 0.0 % (0/3).
- Errores de formulación más frecuentes: `missed_step=12`.
- Ningún item reclamó éxito sobre un fallo (`claimed_success_on_failure=0`).
- Trazabilidad de números (Faith.) media con tools: 100.0 %.
- Costo total 0.2014 USD; el método más caro fue `single_call:cot` con 0.0831 USD (41 %).
- Métodos con tools: 1.00 tool calls medias por item (máximo 1).

6. Archivos

Rutas absolutas bajo `results_validation_gpt-5.4_prompting.report.rescored.json` (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↪ /power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/report.json [leído:
↪ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/scoreboard_per_case.json

```

```
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-
↳ Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/scoreboard_per_case.rescored.json
traces (15 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_prompting/traces
```